

Insulation Condition Monitoring in Distribution Power Grid via IoT-Based Sensing Network

Tianxin Zhuang^{ID}, Ming Ren^{ID}, Xuze Gao, Ming Dong, Wenguang Huang, and Chongxing Zhang

Abstract—Insulation condition monitoring is essential for the safety of our distribution power grid. However, the traditional monitoring solution is hardly applied due to the concerns that arise from natures of distribution grid, such as complex feeder networks, massive operation site, and high economic benefit. In this paper, a wireless sensing network based on Internet of things (IoT) technology is demonstrated for insulation condition perception. First of all, the general framework of the wireless sensing network is designed with considerations on extendibility and accessibility. Next, it goes into detail about the technical requirements of the essential units of the sensing network including sensor node, power management, and data communication. With this concept, two practical cases of IoT-based partial discharge sensing networks realized on the switchgear cabinet and power cable are presented with details involving the design of transient earth voltage and high frequency current sensor nodes, the energy consumption, Lora/NB-IoT-based network structures, and data usage strategies. This IoT-based insulation condition monitoring is proven to be applicable for the distribution grid and expected to be referenced for more possible IoT applications.

Index Terms—Internet of things (IoT), power grid, condition monitoring, WSN, LoRa.

I. INTRODUCTION

INSULATION condition monitoring of power equipment is a powerful way to ensure the safety operation of the power grid [1]. In pursuit of an effective early warning for the potential faults, a host of on-line monitoring approaches are developed to perceive the insulation status information such as temperature rise, partial discharge and leakage current, which are playing the indispensable role in our main power grid in terms of protection, maintenance and asset management [2]. The monitoring system applied to high voltage (HV) power apparatuses normally consists of active sensors, digital-to-analog converter (DAC) units, cable or fiber cable, supervisory control and data acquisition (SCADA) and a variety of data interfaces, whereas for medium

voltage (MV) distribution power network, such complex and costly framework is hardly applicable due to the different concerns arise from the features of distribution grid such as the complex feeder networks, the huge amounts of operation site and the high economic benefit [3]. Besides, the massive heterogeneous and diverse information continuously yielded by the monitoring nodes bring the great challenges for data storage, information matching as well as advanced data usage. According to statistics, China's MV distribution network has 3.08 million kilometers of overhead lines, 610 thousand kilometers of cable, 4.4 million distribution transformers and 4.01 million distribution switch apparatus, and the figures is increasing rapidly. Thus, to explore a more efficient, more reliable as well as more economical solution for distributed monitoring is pressing, in pursuit of our smart distribution grid [4].

Internet of Things is a rapidly developing technology that acquires the information of the physical world by deploying various devices with certain sensing, computing, execution and communication capabilities, and achieves the transmission, coordination and processing of the information through the network to realize the exchange of information between people, things and things [5]. In accordance with the rapid growth of the IoT, low power wide area (LPWA) technologies has been developed and applied to an increasing number of fields not only in consumer market but also industrial circle owing to its lightweight and low-cost framework, long-distance communication, and various schemas for advanced data usage, and thus it is deemed as an ideal framework solution for the insulation status monitoring in distribution grid [6]. However, there are still some challenges that needed to be overcome before the distributed IoT monitoring could be applied [7]. Above all, in the physical level, to develop the extensible wireless sensor network (WSN) with units of low power, low cost and maintenance-free is the basis of reliable sensing layer as well as of effective status perception. Second, a stable running and flexible framework should be designed to ensure the relevant data-driven applications. Beyond that, to promote the transition from traditional distribution grid to intelligent grid, a strategy for the integration of the ubiquitous IoT network and existing electrical communication standard protocols (e.g., IEC 61850) is also essential [8].

In this paper, according to the nature of distribution power grid, a general framework of IoT-based condition monitoring is designed by considering IoT sensor design and distributed WSN, data interaction and data usage (Section II.A). Then the detailed technical requirements of the essential units of the sensing

Manuscript received November 25, 2018; revised March 20, 2019; accepted May 18, 2019. Date of publication May 22, 2019; date of current version July 23, 2019. This work was supported in part by the National Key Research and Development Program of China under Grant 2018YFB0904400 and in part by the National Natural Science Foundation of China under Grants 51877171 and 51507130. Paper no. TPWRD-01351-2018. (Corresponding author: Ming Ren.)

The authors are with the State Key Laboratory of Electrical Insulation for Power Equipment, School of Electrical Engineering, Xi'an Jiaotong University, Xi'an 710049, China (e-mail: ztx1024060195@stu.xjtu.edu.cn; renming@xjtu.edu.cn; gaoux1114@126.com; dongming@xjtu.edu.cn; wghuang@stu.xjtu.edu.cn; zhangcx111@126.com).

Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TPWRD.2019.2918289

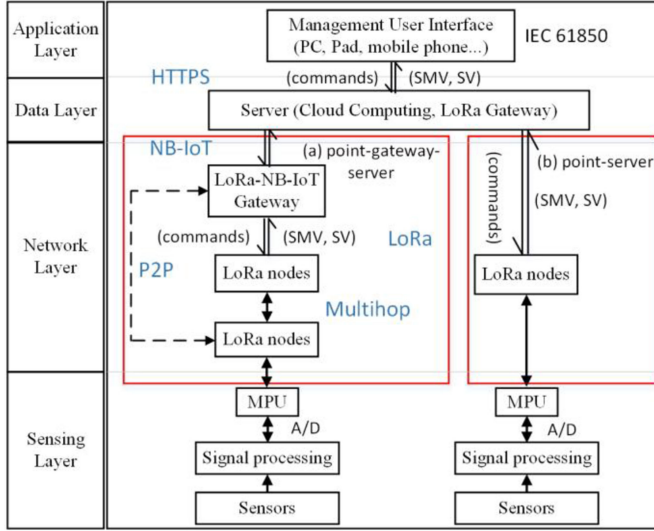


Fig. 1. Framework of the wireless sensing network system.

network including sensor node, data communication and power management is discussed (Section II.B, C). As examples, the IoT-based distributed partial discharge (PD) monitoring is realized on MV switchgears as well as MV cables with technical details in Section III. Based on this, a feasible strategy of fault early warning are demonstrated in Section IV. This work offers a reference implementation for the future IoT-based condition monitoring of the wide range of apparatus serving in distribution grids.

II. FRAMEWORK OF IoT-BASED SENSING NETWORK FOR MV DISTRIBUTED GRID

A. Overall Design of IoT-Based Sensing Network

An general integrated monitoring system should be featured by a good extendibility as well as a good accessibility, and thus it is designed to have four layers, i.e., sensing layer, network layer, data layer and application layer [9], as shown in Fig. 1.

- Sensing layer is the basis of the IoT monitoring, which enables the network to acquire sampled measured values (SMV) and state variables (SV) from sensor nodes and assign new commands to sensor nodes [10]. A distributed sensor node can be a low-power IC or integrated detector depending on the physical object, which normally consists of a sensor, a signal preprocessing module, an analog to digital (A/D) module and a microprocessor unit (MPU).
- The up/down data-link through the sensing layer and data layer is implemented by an IoT-based network layer which is comprised by low-power and wide area network (WAN) transceiver units [11]. In this layer, SMV, SV and commands transmitted in bidirectional point-server or point-gateway-server modes with millisecond delays depending on the distribution area of the monitoring objects (e.g., cable (point-gateway-server), switchgear (point-server) and transformer (point-server)).

- Data layer is driven by a server center for data storage, data management, and advanced data usage and should provide a number of mapping interfaces for responding data access initiated by application layer using standard protocols.
- Application layer [12] contains infrastructure, middleware, operating environment and integration framework, component library, apps, etc. All data access events initiated by other terminals and down data-link command events assigned by an administrator are implemented in this layer via internet.

B. Data Communication Built on LPWA

Considering the practical conditions of distributed monitoring on various apparatus such as the complex interference environment, the large number of sensor site and the wide range of distribution, a reliable low power wide area (LPWA) communication is essential in practical applications and should be tailored to the specific requirements such as data rate, delay, mobility, coverage and so on [13]. For example, PD sensor sites installed on MV cable system are distributed in a wide area of several tens of kilometer determined by the locations of the cable joints or terminals and the distances between them. In this case, the outdoor low-power WAN (LPWAN) technologies are preferred, e.g., eMTC, Nb-IoT, LoRa, Sigfox. For a mass of full-time PD monitoring sites installed on MV switchgear apparatuses which are normally concentrated in one or several rooms, wireless local area networks (LAN), such as Bluetooth, ZigBee and so on [14] are rather appropriate as basic network nodes. With regards to LPWAN application cases, eMTC is easy to deploy and has advantages in terms of mobility and data rate, but its chip and user equipment (UE) are costly, and consume more energy. Sigfox excels in transmission distance and cost, but the deployment is a big challenge, and the throughput (data transfer rate is 100 bps) is too low. LoRa has an excellent power efficiency and supports a ultra-long-distance communication, which is suitable for regional deployment. NB-IoT has advantages of low cost and ultra-wide coverage, and supports quality of service as well as security. Overall, LoRa and NB-IoT are more economical solutions which can be adopted in most cases of monitoring in distribution power grid.

LoRaWAN has an excellent power efficiency (peak current of 32 mA and sleep current of 1 μ A) and supports a long-distance communication up to 40 km in rural areas and 5 km in urban areas. Its unlicensed network (500 kHz–125 KHz) can apply an adaptive modulation technique with multichannel multi-modem transceiver in the base station to receive a multiple number of messages from the channels. The data rate can be managed according to the required data bit rate. Taking PD monitoring as an example, if the sampling rate of the sensor, R , is 1 MS/s (for a frequency-reduced PD pulse with time duration of 10 μ s. See Section III), the maximum density of monitoring, N , can be estimated as 122 /s by (1) [15].

$$N = 1/RT \cdot SF / [2^{SF}/BW] \quad (1)$$

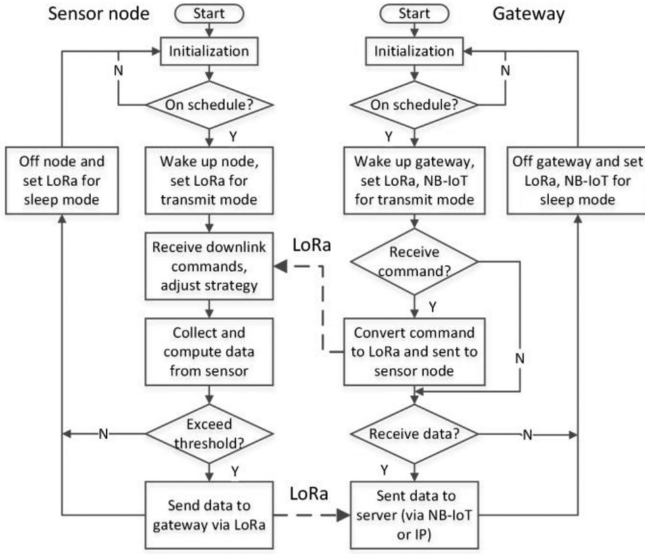


Fig. 2. The flowchart of the communication between sensor nodes and gateways.

where T is the pulse duration time, $10 \mu s$, SF is spreading factor, 10, and BW is modulation bandwidth, 125 kHz. Thus, such LoRaWAN can match the general requirement on monitoring density of PD or other objects and is suitable as a basic nodes connected to sensing layer.

NB-IoT is a new 3GPP wireless access technology with licensed LTE bandwidth (180 KHz) which is derived from GSM and LET technologies. There are 12 subcarriers of 15 kHz in downlink using OFDM and 3.75/15 kHz in uplink using SC-FDMA, which provides a great potential for high speed multichannel data transmission [16]. Utilized with Lora node substrate, the network layer can be built with good extendibility and interactivity. The flow chart of the communication between sensor nodes and gateways is shown in Fig. 2. When the communication starts, the node and gateway is wakened up, and the LoRa and NB-IoT is set to transmit mode. Only then can the gateway send commands and the sensor node receive commands. During the monitoring period, the sensor node acquires the SMV and SV, and if the SMV exceed the threshold, the data will be send to the gateway via LoRa, otherwise the node and gateway will be set to sleep state after monitoring.

C. Basic Principles in IoT Sensor Design

As the basis of IoT monitoring network, the wireless sensor is expected to have a service life as long as the monitoring object and a long battery life with guarantee of data reliability. Beyond that, the sensor nodes deployed on site are normally required to be small, weight and cheap [17]. Based on these requirements, a wide range of commercial micro integrated sensors and MEMS IC sensors can be directly employed for various status monitoring such as temperature, humidity, vibration and so on. However, with regards to the edge characterization objects or other advanced functionalities that need to be performed by load cell for insulation condition monitoring (e.g., PD activity, leakage

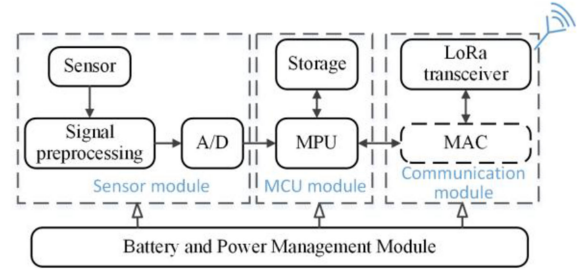


Fig. 3. The structure block diagram of the wireless sensor node.

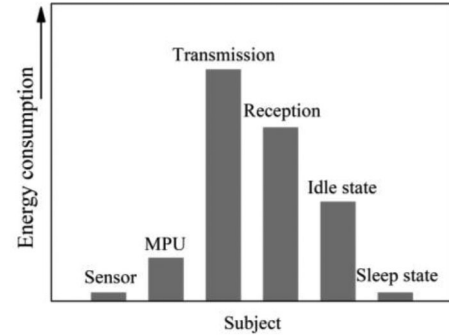


Fig. 4. Typical power consumption of each subject of the sensor node.

current and some time-resolved status parameters), more fundamental design of sensor should be carried out to support the refined diagnosis. In this case, the signal coupler, preprocessing, processor, power supply and data transceiver should be matched up to pursue a balance among data precision, timeliness and battery life. A general structure block diagram of the wireless sensor node is shown in Fig. 3.

Since a mass of nodes are distributed in a wide area, the battery replacement should be possibly avoided. Hence, an efficiency power management and refined operation strategy are the key premises for the design of the sensor node [18]. In the framework above, the typical energy consumption of each unit of the sensor node is shown in Fig. 4. Obviously, the transmission part and reception part dominate the consumption. Besides, the node also consumes quite a bit of energy even in an idle state.

To reduce energy consumption, a dynamic handoff scheme among active, idle and sleep modes of MPU should be introduced properly. Beyond that, the economization on computing power, the queue cache storage as well as data transfer size are also expected [19], which means the sampling rate and resolution of the data acquisition have to be reduced in the premise of acceptable data effectiveness. In the case of the rapid event acquisition (e.g., PD, overvoltage, vibration, etc.), the original responses of coupler should be converted to a demodulated signals with narrow bandwidth before sent to the A/D converter. Taking an advanced RISC machine (ARM, 1MS/s) as an example, if the upper cutoff frequency of the coupled signal exceeds 500 kHz, the down-conversion is indispensable for ensuring the sampling accuracy. Besides the energy consumed by the processor, more energy is used for the data transmission and reception. To reduce the energy consumption of data transmission, the data

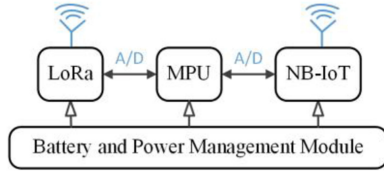


Fig. 5. The structure block diagram of the LoRa-NB-IoT gateway.

bit rate and detecting density of the sensor node should be adjusted at a relatively low level [20].

As aforementioned, considering the extendibility and the accessibility of the sensing network, a framework featured by NB-IoT substrate with LoRa-gateway is preferred as shown in Fig. 5, which can achieve LoRa communication with sensor nodes and NB-IoT communication with base station (or cloud). Based on that, not only the collection of the information of sensor nodes but also the conversion between LoRa and NB-IoT can be realized easily.

Overall, the core requirements on building a IoT-based sensing network are:

- The sensor should be guaranteed to have low power consumption, low cost and digital integration.
- Data processing should be performed in sensing layer with lean and uniform structures in pursuit of low power consumption transmission.
- Data transmission network should be of strong scalability, stability and flexibility.

III. APPLYING IoT-BASED CONDITION MONITORING IN MV DISTRIBUTION GRID: CABLE AND SWITCHGEAR

Based on the framework of IoT-based sensing network, two practical application cases are demonstrated in this section. First, PD monitoring on MV cable which is the most widespread equipment in a MV grid. Second, PD monitoring on massed switchgear cabinets which are the typical indoor applications of multipoint monitoring. Both of the two cases need deploying PD sensor nodes in a wide area of several tens of kilometers with interlinked network, which can hardly be realized by traditional on-line monitoring because of the unendurable cost and complicated engineering reconstruction.

A. PD Sensor Developed for IoT Network

PD detection is a popular approach for insulation condition monitoring of power apparatus and a range of PD coupling methods are developed based on the different associated physic phenomena such as current pulse flow, electromagnetic (EM) wave and acoustic propagation, among which transient earth voltage (TEV) method and high-frequency current (HF) method are the most common ones applied on switchgear cabinets and MV cables respectively [21], [22]. Based on the coupling principles of TEV and HF, the sensor nodes for switchgears and for cables are designed with the general structure described following, which contains a coupler module, a preprocessing module, a processor

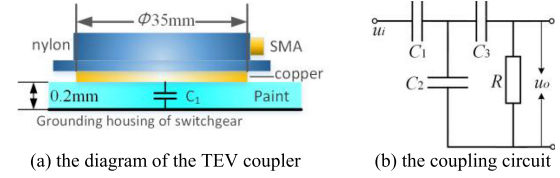


Fig. 6. The design of the TEV coupler.

module, a communication module, a power supply and a power management module. Since the original signals of PDs are characterized by fast rise time, short duration and high repetition rate, a careful design of signal down-conversion is indispensable to decrease the power consumption and hardware cost of MPU as well as data storage.

1) *TEV Sensor Node for Switchgear*: The capacitive TEV coupler is used in the sensor module as shown in Fig. 6a. It can couple the PD-induced transient voltage signals that leak from non-complete EM shielding part and pass on the surface of the grounding metal of a switchgear cabinet. With the capacitance generated between the metal casing and the circular copper plate of the sensor (C_1), the filter capacitance (C_2), the coupling capacitor (C_3) and the output resistance (R), the simple coupling circuit can be drawn as Fig. 6b where u_i and u_o refer to the PD-induced transient voltage and the coupled signal by the sensor, respectively. As the thickness of the paint is about 200 μm according to the production standards [23] of switchgear cabinet, C_1 can be estimated as 135 pF by using (2).

$$C_1 = \epsilon S / d \quad (2)$$

where S is the coupling area, ϵ is the permittivity, d is the thickness of the paint.

Considering that C_1 is much greater than C_2 , the transfer function of the coupling circuit can be given in (3). To obtain a better sensitivity, the value of R should be chosen much greater than $(1+1/k)/\omega C_1$ (where the ω is the angular frequency of the signal). In particular, as C_2 and C_3 are chosen as 1 pF and 150 pF respectively, R should be much greater than 22 Ω (in our case, R is chosen as 5 k Ω) to ensure an acceptable sensitivity in coupling the TEV signals with a main frequency below 100 MHz. Finally, the main frequency point and bandwidth of the coupler are determined as 225 MHz and 450 MHz respectively by using a spectrum analyzer.

$$T(s) = [1 + (1 + k) / k C_1 R s]^{-1} \quad (3)$$

where k is the ratio of C_3 to C_1 .

To pursue a wide pulse width, a good interference immunity as well as a wide dynamic range, the preprocessing should include a bandpass filter (with bandwidth ranging from 1 M to 30 MHz to exclude the EM noise), a low-noise amplifier (with adjustable gain between 10 and 500 times), and a demodulation (175 kHz), as shown in Fig. 7.

Fig. 8 shows the TEV PD sequences with and without the preprocessing which demonstrates the effectiveness of the preprocessing approach on reduction of signal frequency and

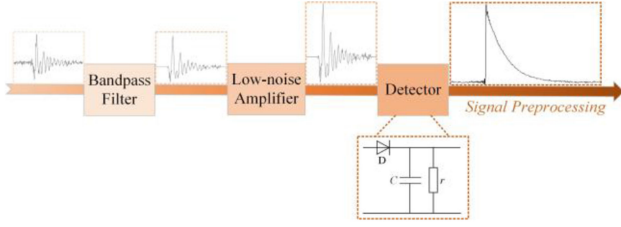


Fig. 7. The signal preprocessing of the TEV signal.

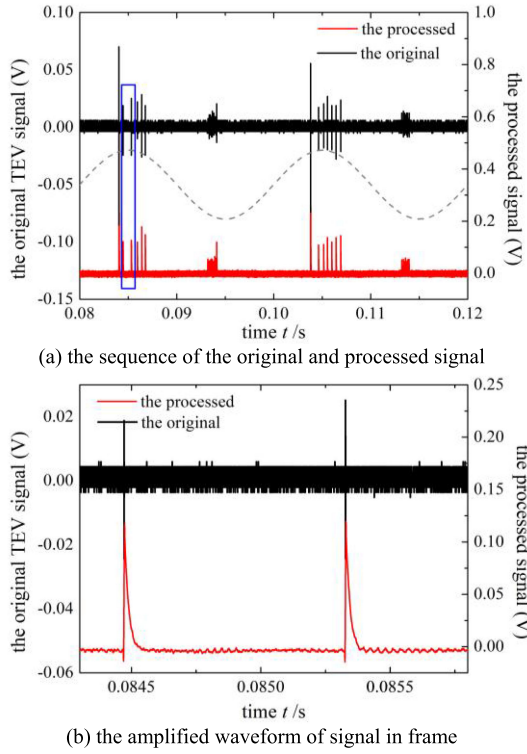


Fig. 8. The sequence and the amplified waveform of the original signal and the processed signal.

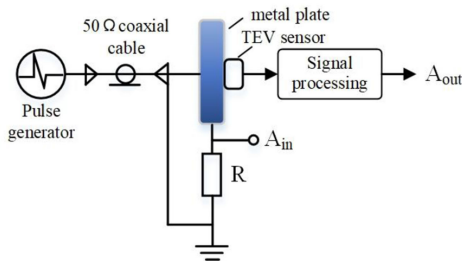


Fig. 9. The calibration test of the TEV sensor module.

improvement of signal-to-noise ratio(SNR). The rise time of the processed signal (t_r) is about $2 \mu\text{s}$ and its bandwidth is about 175 kHz ($\text{BW} = 0.35/t_r$). Therefore, an A/D module with 400 kHz sampling rate is sufficient to handle with the processed TEV signal below 350 kHz .

To provide a criterion for monitoring, the TEV sensor module is calibrated by experiment shown in Fig. 9. A Gaussian pulse with rise time of 12 ns (approximately 30 MHz) and amplitude

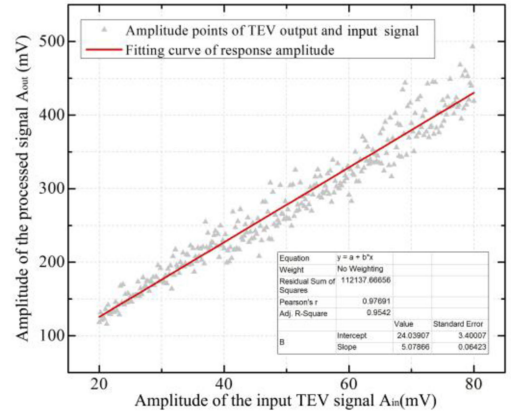


Fig. 10. The calibration results of TEV sensor module.

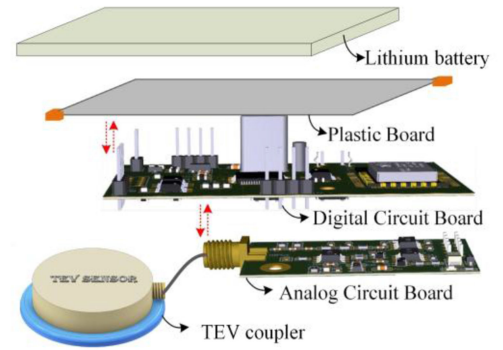


Fig. 11. The actual structure of the sensor node.

ranging from 20 mV to 80 mV is employed as the calibration input and applied to a metal plate through a 50Ω coaxial cable. A standard 50Ω impedance (R) is used to measure the apparent TEV amplitude. The fitting results of correspondence between the amplitudes of the processed and the apparent signals is presented in Fig. 10 which follows a linear relationship as:

$$A_{\text{out}} = 24.04 \text{ mV} + 5.08 \times A_{\text{in}}, A_{\text{in}} \in [20 \text{ mV}, 80 \text{ mV}] \quad (4)$$

where A_{out} (mV) and A_{in} (mV) refer to the amplitudes of the processed and the apparent signals respectively. In practical use, the strength of TEV signal is ultimately expressed in the unit of dBmV with a conversion of (5), where P is the logarithmic form of the strength of TEV.

$$P = 20 \log_{10} ((A_{\text{out}} - 24.04 \text{ mV}) / 5.08) \quad (5)$$

The actual structure of the sensor node is shown in Fig. 11.

2) *HF Sensor Node for Cable*: The integrated sensor developed for PD monitoring of power cable is shown in Fig. 12, which contains a SNR-improved Rogowski coil (bandwidth 800 kHz – 106 MHz , sensitivity 1 V/A in band), a signal preprocessing circuit, a MPU module (in our case a 120 MHz ARM is employed) and a wireless transceiver. The structure and equivalent circuit of the Rogowski coil are shown in Fig. 13(a) and Fig. 13(b), where two loops are wound symmetrically on the core to double the signal and cancel the noise. The relative bandwidth

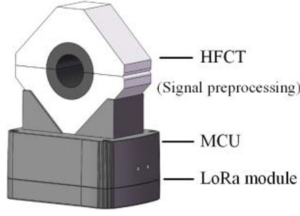


Fig. 12. The sensor node for cable.

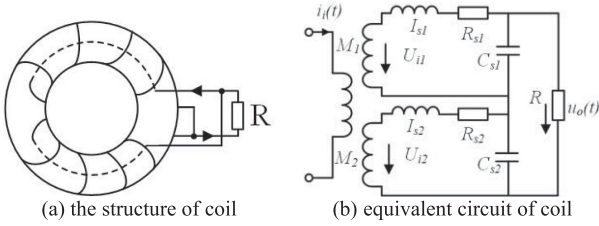


Fig. 13. The structure and equivalent circuit of SNR-improved Rogowski coil.

of the coil is determined by

$$f_{\max}/f_{\min} \propto \mu l h \ln(d_2/d_1) / (d_2 - d_1 + 2h^2) \quad (6)$$

where h , d_1 , d_2 refer to the height, the inner and outer diameter of the core, l refers to the total length of the core. It is obvious that the relative bandwidth reaches the maximum when $h = (d_2 - d_1)/2$. In this work, h , d_1 , d_2 are chosen 8, 48, 64 mm respectively.

For the case of cable, the Lora multihop mode is preferred rather than Lora point-to-point mode because that the sensor nodes installed on cables are distributed in a relatively wider range comparing to the sites of TEV sensor nodes for switchgear apparatus). Such high-frequency current (HF) IoT sensors can be installed at the ground leads drawn from the cable connectors or terminals, forming a fault-traceable network as sensor nodes themselves are location information carriers.

The frequency component of the discharge pulse current coupled by the Rogowski coil is ranging from dozens to hundreds MHz, so the design of the signal preprocessing is similar to the first case, which includes the filtering, amplification and demodulation. To determine the response of HF sensor node, a standard pulse current generator is employed as a calibrator which can produce repeated current pulses with rise time of 5 ns and apparent charge of 5~500 pC, and the output of the precision resistor connected in series with the test capacitor is used as the baseline. The relationship between the amplitude of the output of the sensor node and the apparent charge is fitted well by (7), where q_a (pC) is the apparent charge, A_{out} (V) is the amplitude of sensor output.

$$q_a = 298 \text{ pC/V} \times A_{out} \quad (7)$$

B. Data Transmission

1) *The Data Transmission in Switchgear PD Monitoring:* Since the switchgear cabinets are distributed in a centralized region, normally an indoor substation or an energy-intense factory, it is convenient to set one gateway within the reaches of



Fig. 14. The point to server communication applied in an electrolysis plant. Sensor nodes are deployed at the switchgear in 10kV switch houses, and the up/down data are transmitted via the links between LoRa nodes and LoRa gateway set in the control room.

all sensor nodes with LoRa connection. In this case, the point-to-sever communication (as shown in Fig. 1b) is adopted in the switchgear PD monitoring. The data transmission route in an electrolysis plant is shown in Fig. 14.

79 TEV sensor nodes are installed in three 10 kV switch houses; a LoRa gateway connected to a server is set in the control room with a distance less than 1km away from the farthest node. A cloud database will be updated synchronously during each event of data upload.

The up/down data between the nodes and sever communicate directly through LoRa in the active period of the sensor. In the uplink data transmission, the only the brief SMV processed by MPU of the sensor instead of real-time waveform is sent with SV information to gateway at a bit rate about 20 Kbs.

- SMV: max. & mean PD quantities/repetition rate/test duration.
- SV: timestamp/time interval/battery level/object ID/sensor ID/error code/bit rate

2) *The Data Transmission in Cable PD Monitoring:* Due to the ultra-large span of the cable lines, the distances among the monitoring points may be as long as several tens of kilometers. Therefore, the data communication in this case is implemented by combining the mode of LoRa multihop between the nodes and the mode of LoRa point-to-gateway, as the data routes shown in Fig. 15. In this case, the gateways uses NB-IoT to communicate with the cloud server. Five sensor nodes are deployed at the terminals and connectors of the cable.

C. Energy Consumption

The energy consumption of the designed sensor node in each state is estimated by that of the main power consumption parts, including MPU, amplifier, the transmission and reception of the LoRa module. The assessed results of these components and the tested results of the overall node are listed in Table I, which are calculated by multiplying the nominal and measured currents and the power supply voltages. Based on the tested energy



Fig. 15. The point-gateway-sever (cloud) communication in the power cable PD monitoring in Jinan city of China. Sensor nodes are deployed at the terminals and connectors, and the communication between nodes and cloud is passed by LoRa-NB-IoT gateway.

TABLE I
ASSESSMENT AND MEASUREMENT OF ENERGY CONSUMPTION UNIT: mW

	Active state	Idle state	Sleep state
MPU	2.5	0.05	0.01
Amplifier	25	0	0
Transmission	105	0.006	0.006
Reception	90	45	0.006
Estimated	222.5	45.056	0.022
Measured	240	51	0.04

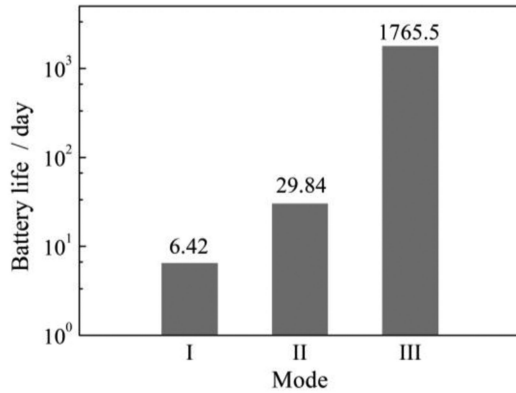


Fig. 16. The battery (3.7 V, 5000 mAh) life of three operation modes. Mode I: full time of active state. Mode II: 5 minutes of active state+55 minutes of idle state for each hour. Mode III: 5 minutes of active state+55 minutes of sleep state for each hour.

consumptions, the life of a 3.7 V, 10000 mA·h battery in three modes are calculated, the results are shown in the Fig. 16. It is seen that the battery life in the mode III (for 5 minutes daily of active state and other time of sleep state) can be up to 5 years.

At present, operating and maintenance (in China) of cable mainly rely on the one year-cycled live detection and the 3~5 years-cycled routine testing. Taking a municipal- scale regional power grid as an example, it needs about 200 persons per year

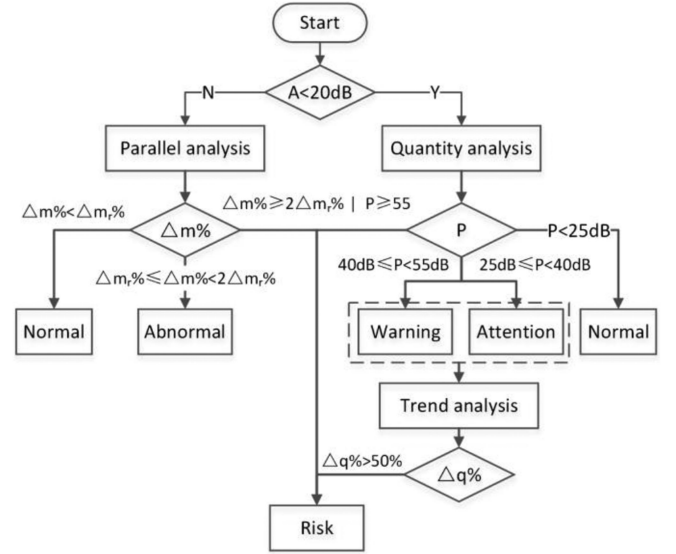


Fig. 17. The flowchart of dynamic threshold assessment strategy on switchgear.

for the live detection and routine testing and 5 hours each time, thus the total labor costs are 10000 dollars per year (according to the current hourly rate in China, normally 10 dollars per hour). If the distributed sensors are deployed for one sensor per km in a 200 km length of cable system, according to the battery life of 5 years, 3 dollars per battery, and 20 minutes of replacement per sensor (traffic time is considered), the total costs are about 254 dollars per year.

IV. DIAGNOSIS STRATEGY BASED ON IOT MONITORING

A. Status Assessment Strategy on Switchgear

Partial discharge detection and diagnosis is a critical technical means to find the sudden as well as the developing insulation defects, and plays an important role in assessing the insulation state of power equipment. Magnitude, frequency, phase-resolved statistics of PD are important characteristic parameters to reflect the PD severity as well as fault pattern. The relationship of PD event to the phase voltage waveform can help to recognize the PD pattern in the in-depth status diagnosis, but the diagnosis has a relatively high requirement on the computing rate and memory of the processor which also gives rise to a high power consumption. In this work, we focuses on the construction of the wireless sensor network framework rather than the diagnostic method of PD, so only a simple status assessment strategy is demonstrated for data usage. Taking the strategy on the switchgear as an example, the flow chart of the assessment strategy is shown in Fig. 17, where A means the value of the amplitude information, P is the logarithmic form ($\sim$ dB) of the strength of TEV, $\Delta m\%$ is the deviation of the target switchgear, and $\Delta q\%$ is the variation of P with time, which are explained follow.

1) *Quantity Analysis*: In the assessment, the status of the switchgear can be divided into four levels of normal, attention, warning and risk, based on the empirical values derived from the

TABLE II
THE CRITERION FOR QUANTITY ANALYSIS UNIT: dB

Criterion	Basic status	Treatment
$P \leq 25$	Normal	Routinization
$25 < P \leq 40$	Attention	Shorten monitoring interval
$40 < P \leq 55$	Warning	Shorten monitoring interval and upload results to host PD
$P > 55$	Risk	Upload results to host PD and turn on real-time monitoring mode

TABLE III
THE CRITERION FOR PARALLEL ANALYSIS

Criterion	Basic status
$\Delta m\% \leq \Delta m_r\%$	Normal
$\Delta m_r\% < \Delta m\% \leq 1.5\Delta m_r\%$	Attention
$1.5\Delta m_r\% < \Delta m\% \leq 2\Delta m_r\%$	Warning
$\Delta m\% > 2\Delta m_r\%$	Risk

statistics of the experimental data and failure cases. The quantity analysis uses P as the criterion to directly judge the status, which is listed in Table II. It provides the basic status of the switchgear without considering the interference and background noise.

2) *Trend Analysis*: When the status is judged as attention or warning status by the quantity analysis, it needs to increase the monitoring density and intensively monitor the trend over time, i.e., the trend analysis. It uses an indirect value calculated as the relative variation of the amplitude information with time as given follow:

$$\Delta q\% = (A_{t2} - A_{t1}) / (A_{t1} \cdot \Delta t) \times 100\% \quad (8)$$

where A_{t1} and A_{t2} are the amplitudes before and after a time interval Δt . The time interval can be adjusted automatically according to the status results or manually when the start time comes. The trend analysis determine whether the insulation status is the developing deterioration or sudden fault. If the variation exceeds 50%, the status is assessed at risk.

3) *Parallel Analysis*: When it comes to the same type of switchgear in the same environment, it can use parallel analysis to determine the fault location and severity. To characterize the differences between the target apparatus and the others, the deviation from average $\Delta m\%$ is calculated by the (9).

$$A_{av} = k^{-1} \sum_{n=1}^k A$$

$$\Delta m\% = (A - A_{av}) / A_{av} \times 100\% \quad (9)$$

where A is the amplitude information of each switchgear, A_{av} is the average level of the apparatuses.

A reference value $\Delta m_r\%$ is set to be a criterion in the parallel analysis as shown in Table III, where the $\Delta m_r\%$ is given by the experiment.

B. Dynamic Monitoring Strategy and Fault Location

As mentioned above, the monitoring density needs to vary with the status to realize the refined diagnosis. When there is something abnormal in the status according to the quantity analysis, trend analysis and parallel analysis, the server will automatically send a presupposed command to shorten the monitoring interval to 1 hour or shorter. Also, the user can use the command to change the start time in the active period although no abnormal state is detected.

Moreover, the location information is recorded in the node storage and sent together to the server in the monitoring. Since the monitoring nodes in both cases are not mobile, it can achieve the fault location (equipment-level location, e.g., which switchgear or cable joint is faulty) by matching the location information of the sensor nodes without the additional GPS hardware.

V. CONCLUSION

In this paper, a wireless sensing network based on IoT technology is demonstrated in pursuit of an efficiency condition monitoring solution for MV distribution power grid. Firstly, a general framework of the wireless sensing network including sensing layer, network layer, data layer and application layer is designed for extendibility and accessibility. Based on that, the technical requirements on the essential units of the IoT sensing network including integrated sensor node, power management and LPWA communication are discussed in detail. With the propose framework, two practical cases of IoT-based partial discharge sensing networks realized on the switchgear cabinet and power cable respectively are presented with the details involving the design of TEV and HF sensor nodes, the energy consumption, Lora/NB-IoT-based network structures and data usage strategies. Comparing to the traditional monitoring solution, such IoT-based distributed insulation condition monitoring is proven more applicable for the distribution grid and expected to be references for more possible IoT applications.

REFERENCES

- [1] C. R. Li *et al.*, "Condition monitoring and diagnosis of high-voltage equipment in China-recent progress," *IEEE Elect. Insul. Mag.*, vol. 29, no. 5, pp. 71–78, Sep.–Oct. 2013.
- [2] S. Ceferin *et al.*, "Power transformer monitoring systems for better asset management," in *Proc. Int. Conf. Exhib. Elect. Distrib. (CIRED)*, Glasgow, U.K., 2017, pp. 395–399.
- [3] A. Abdrabou, "A wireless communication architecture for smart grid distribution networks," *IEEE Syst. J.*, vol. 10, no. 1, pp. 251–261, Mar. 2014.
- [4] M. Y. Zhai, "Transmission characteristics of low-voltage distribution networks in China under the smart grids environment," *IEEE Trans. Power Del.*, vol. 26, no. 1, pp. 173–180, Jan. 2011.
- [5] S. E. Bibri, "The IoT for smart sustainable cities of the future: An analytical framework for sensor-based big data applications for environmental sustainability," *Sustain. Cities Soc.*, vol. 38, pp. 230–253, 2017.
- [6] X. Wang and P. Yi, "Security framework for wireless communications in smart distribution grid," *IEEE Trans. Smart Grid*, vol. 2, no. 4, pp. 809–818, Dec. 2011.
- [7] M. T. Lazarescu, "Design of a WSN Platform for long-term environmental monitoring for IoT applications," *IEEE. J. Emerg. Sel. Topics Circuits Syst.*, vol. 3, no. 1, pp. 45–54, Mar. 2013.
- [8] T. S. Sidhu and Y. Yin, "Modelling and simulation for performance evaluation of IEC61850-based substation communication systems," *IEEE Trans. Power Del.*, vol. 22, no. 3, pp. 1482–1489, Jul. 2007.

- [9] M. Han, L. Cheng, and H. Meng, "Application of four-layer neural network on information extraction," *Neural Netw.*, vol. 16, no. 5, pp. 547–553, Aug. 2003.
- [10] W. He, G. Yan, and L. D. Xu, "Developing vehicular data cloud services in the IoT environment," *IEEE Trans. Ind. Inform.*, vol. 10, no. 2, pp. 1587–1595, May 2014.
- [11] M. Centenaro *et al.*, "Long-range communications in unlicensed bands: The rising stars in the IoT and smart city scenarios," *IEEE Wireless Commun.*, vol. 23, no. 5, pp. 60–67, Oct. 2016.
- [12] J. Yick, B. Mukherjee, and D. Ghosal, "Wireless sensor network survey," *Comput. Networks*, vol. 52, no. 12, pp. 2292–2330, 2006.
- [13] K. Zheng *et al.*, "Design and implementation of LPWA-based air quality monitoring system," *IEEE Access*, vol. 4, pp. 3238–3245, Jun. 2016.
- [14] S. Y. Shin *et al.*, "Packet error rate analysis of zigbee under WLAN and bluetooth interferences," *IEEE Trans. Wireless Commun.*, vol. 6, no. 8, pp. 2825–2830, Aug. 2007.
- [15] R. S. Sinha, Y. Wei, and S. H. Hwang, "A survey on LPWA technology: LoRa and NB-IoT," *ICT Express*, vol. 3, pp. 14–21, 2017.
- [16] Z. Sheng *et al.*, "Recent advances in industrial wireless sensor networks toward efficient management in IoT," *IEEE Access*, vol. 3, pp. 622–637, May 2015.
- [17] K. Latif *et al.*, "Energy consumption model for density controlled divide-and-rule scheme for energy efficient routing in wireless sensor networks," *Int. J. Ad Hoc Ubiquitous Comput.*, vol. 21, no. 2, pp. 130–139, 2016.
- [18] M. Dong, K. Ota, and A. Liu, "RMER: Reliable and energy-efficient data collection for large-scale wireless sensor networks," *IEEE Internet Things J.*, vol. 3, no. 4, pp. 511–519, Aug. 2017.
- [19] J. Ren, Y. Zhang, K. Zhang, A. Liu, J. Chen, and X. S. Shen, "Lifetime and energy hole evolution analysis in data-gathering wireless sensor networks," *IEEE Trans. Ind. Inform.*, vol. 12, no. 2, pp. 788–800, Apr. 2016.
- [20] D. Xie, W. Wei, Y. Wang, and H. Zhu, "Tradeoff between throughput and energy consumption in multirate wireless sensor networks," *IEEE Sens. J.*, vol. 13, no. 10, pp. 3667–3676, Oct. 2013.
- [21] M. Ren, M. Dong, Z. Ren, H.-D. Peng, and A.-C. Qiu, "Transient earth voltage measurement in PD detection of artificial defect models in SF₆," *IEEE Trans. Plasma Sci.*, vol. 40, no. 8, pp. 2002–2008, Aug. 2012.
- [22] Y. Xu *et al.*, "Special requirements of high frequency current transformers in the on-line detection of partial discharges in power cables," *IEEE. Elect. Insul. Mag.*, vol. 32, no. 6, pp. 8–19, Nov.–Dec. 2016.
- [23] "Empty enclosures for low-voltage switchgear and controlgear assemblies - General requirements," IEC Standard 62208, 2011.



Tianxin Zhuang was born in Fujian Province, China, in 1995. He received the bachelor's degree in electrical engineering from Xi'an Jiaotong University, Xi'an, China, in 2017. He is currently working toward the graduate degree with HV Technology Institute, Xi'an Jiaotong University.

His major is high voltage engineering and his main research interests are the partial discharge mechanism of both gas and solid dielectrics and the insulation status assessment technology.

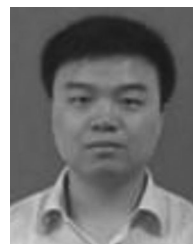


Ming Ren was born in Shaanxi Province, China, in 1987. He received the B.S. and Ph.D. degrees in electrical engineering from Xi'an Jiaotong University, Xi'an, China, in 2009 and 2013, respectively.

He was with Xi'an Jiaotong University as a Lecturer in May 2014, and currently, he is an Associated Professor with HV Technology Institute, School of Electrical Engineering, Xi'an Jiaotong University. His major is high voltage engineering and his present research works are risk assessment of UHV/EHV power equipment, insulation status diagnosis, and advanced fault detection and protection technologies for power system.



Xuze Gao received the bachelor's degree from College of Electrical Engineering and Information Technology, Sichuan University, Chengdu, China. He is currently studying in HV Technology Institute, Xi'an Jiaotong University, Xi'an, China. His major is insulation condition assessment of electrical equipment.



Ming Dong was born in Shaanxi Province, China, in 1978. He received the B.S. and Ph.D. degrees in electrical engineering from Xi'an Jiaotong University, Xi'an, China, in 2001 and 2006, respectively. He was with Xi'an Jiaotong University as a Lecturer in 2007, and currently, he works as a Professor with HV Technology Institute, School of Electrical Engineering, Xi'an Jiaotong University. His major study is aging mechanism of dielectric material and his research work is intelligent HV equipment and power condition assessment.



Wenguang Huang was born in Anhui Province, China, in 1992. He received the bachelor's degree in electrical engineering from Xiamen University, Xiamen, China in 2016. He is currently working toward the graduate degree with HV Technology Institute, Xi'an Jiaotong University, Xi'an, China. His major is high voltage engineering and his main research interests are fault diagnosis and state maintenance of power equipment and the design of radio frequency energy harvesting system.



Chongxing Zhang was born in Shaanxi Province, China, in 1990. He received the B.S. degree in electrical engineering from Xi'an Jiaotong University, Xi'an, China, where he is currently working toward the Ph.D. degree. Currently, he is specializing in the insulation design and assessment of HV/MV power electronic converters. His major is high-voltage engineering and his research interests include gaseous discharge, PD detection and diagnosis, and insulation status assessment of power equipment.